# hdoc day 29 by trishil chauhan

## Aim

Write a menu-driven C program to count even, odd, positive, negative and zero elements; perform linear search; merge two arrays; and exit.

## C Program

```
#include <stdio.h>
#define MAX 100

void readArray(int a[], int *n) {
    int i;
    printf("Enter number of elements (1-100): ");
    scanf("%d", n);
    if (*n < 1 || *n > MAX) {
        printf("Invalid size.\n");
        *n = 0;
        return;
    }
    printf("Enter %d elements: ", *n);
    for (i = 0; i < *n; i++) scanf("%d", &a[i]);
}

int main(void) {
    int a[MAX], b[MAX], merged[2 * MAX];
    int n = 0, m, choice, i, key, found;
    int even, odd, positive, negative, zero;

    do {
        printf("\n--- ARRAY OPERATIONS ---\n");
        printf("1. Count even, odd, positive, negative, zero\n");
        printf("2. Linear search\n");
        printf("3. Merge two arrays\n");
        printf("4. Exit\nEnter choice: ");
        if (scanf("%d", &choice) != 1) return 1;

        switch (choice) {
        case 1:
            readArray(a, &n);
            if (n == 0) break;
            even = odd = positive = negative = zero = 0;
            for (i = 0; i < n; i++) {
                if (a[i] % 2 == 0) even++;
                else odd++;
                if (a[i] > 0) positive++;
                else if (a[i] < 0) negative++;
                else zero++;
            }
            printf("Even: %d, Odd: %d\n", even, odd);
            printf("Positive: %d, Negative: %d, Zero: %d\n",
                   positive, negative, zero);
            break;
        case 2:
            readArray(a, &n);
            if (n == 0) break;
            printf("Enter element to search: ");
            scanf("%d", &key);
            found = 0;
            for (i = 0; i < n; i++) {
                if (a[i] == key) {
                    printf("Found at position %d\n", i + 1);
                    found = 1;
```

```
                    break;
                }
            }
            if (!found) printf("Element not found.\n");
            break;
        case 3:
            printf("First array:\n");
            readArray(a, &n);
            if (n == 0) break;
            printf("Second array:\n");
            readArray(b, &m);
            if (m == 0) break;
            for (i = 0; i < n; i++) merged[i] = a[i];
            for (i = 0; i < m; i++) merged[n + i] = b[i];
            printf("Merged array: ");
            for (i = 0; i < n + m; i++)
                printf("%d ", merged[i]);
            printf("\n");
            break;
        case 4:
            printf("Exiting program.\n");
            break;
        default:
            printf("Invalid choice. Try again.\n");
        }
    } while (choice != 4);
    return 0;
}
```

## Sample Run

```
Enter choice: 1
Enter number of elements (1-100): 5
Enter 5 elements: -2 0 3 4 -5
Even: 3, Odd: 2
Positive: 2, Negative: 2, Zero: 1

Enter choice: 2
Enter number of elements (1-100): 4
Enter 4 elements: 8 6 2 9
Enter element to search: 2
Found at position 3

Enter choice: 3
First array:
Enter number of elements (1-100): 2
Enter 2 elements: 1 2
Second array:
Enter number of elements (1-100): 3
Enter 3 elements: 3 4 5
Merged array: 1 2 3 4 5

Enter choice: 4
Exiting program.
```

## Result

The menu-driven program performs all four operations successfully. Zero is counted as even, but neither positive nor negative.